Chapter 9. Fourier Series

Orthonormal systems, pointwise convergence, Fejér's theorem, Parseval's identity, smoothness and decay

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9.1 Fourier coefficients

In an inner product space X, a family $\{\phi_n : n \in \mathbb{Z}\}$ is **orthonormal** if $\langle \phi_m, \phi_n \rangle = \delta_{mn}$. In $\mathcal{R}^2[-\pi,\pi]$ with $\langle f, g\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} f\bar{g}$, the functions $e_n(x) = e^{inx}$ are orthonormal. For $f \in \mathcal{R}^1[-\pi,\pi]$ (extended 2π-periodically) the **Fourier coefficients** are $\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-int}dt$, the **Fourier series** is $\sum \hat{f}(n)e^{inx}$, and $s_n(f;x) = \sum_{|k|\le n} \hat{f}(k)e^{ikx}$.

Theorem 9.1 (9.1.1, best approximation and Bessel's inequality)

Let $\{\phi_n\}$ be orthonormal in X, $x \in X$, $c_n = \langle x, \phi_n\rangle$. For all complex z_n,

$$\Big\|x - \sum_{n=-N}^{N} c_n\phi_n\Big\| \le \Big\|x - \sum_{n=-N}^{N} z_n\phi_n\Big\|,$$

with equality iff $z_n = c_n$ for all $|n| \le N$. Moreover $\sum_{n\in\mathbb{Z}} |c_n|^2 \le \|x\|^2$.

Proof. Let $s = \sum_{|n|\le N} c_n\phi_n$ and $t = \sum z_n\phi_n$. Then $\langle x-s, \phi_n\rangle = 0$ for $|n| \le N$, so $x - s \perp s - t$, and $\|x-t\|^2 = \|x-s\|^2 + \|s-t\|^2 = \|x-s\|^2 + \sum|c_n - z_n|^2$. With $t = 0$: $\|x\|^2 = \|x-s\|^2 + \sum_{|n|\le N}|c_n|^2$, which gives Bessel's inequality. ∎

Proposition 9.2 (9.1.2)

For $f \in \mathcal{R}^1[-\pi,\pi]$:
(a) if f is real, $\hat{f}(-n) = \overline{\hat{f}(n)}$;
(b) if f is even, $\hat{f}(-n) = \hat{f}(n)$;
(c) if f is odd, $\hat{f}(-n) = -\hat{f}(n)$.

Proof. (a) Conjugate the integral. (b), (c) Substitute $t \mapsto -t$. ∎

In real form, $s_n(f;x) = \frac{a_0}{2} + \sum_{k\le n}(a_k\cos kx + b_k \sin kx)$ with $a_k = \frac{1}{\pi}\int f(t)\cos kt\,dt$, $b_k = \frac{1}{\pi}\int f(t)\sin kt\,dt$; even functions have cosine series and odd functions sine series.

Example 9.1

$f(x) = x$ on $(-\pi,\pi)$: $\hat{f}(n) = \frac{(-1)^n i}{n}$ $(n \ne 0)$, so $x \sim 2\sum(-1)^{n+1}\frac{\sin nx}{n}$. $f(x) = |x|$: $|x| \sim \frac{\pi}{2} - \frac{4}{\pi}\sum_{k \text{ odd}} \frac{\cos kx}{k^2}$, and at $x = 0$ this gives $\sum_{k \text{ odd}} k^{-2} = \pi^2/8$, hence $\sum n^{-2} = \pi^2/6$ (once convergence to f is proved, Corollary 9.7).

9.2 Pointwise convergence

Theorem 9.3 (9.2.1, Riemann–Lebesgue lemma)

If $\int_I |f| < \infty$ on an interval I, then for every $\beta \in \mathbb{R}$, $\lim_{\alpha\to\infty}\int_I f(t)\sin(\alpha t + \beta)\,dt = 0$.

Proof. For $f = \chi_{[a,b]}$, $|\int_a^b \sin(\alpha t + \beta)dt| \le 2/\alpha$; by linearity the claim holds for step functions. For general f and ϵ, choose a step function s with $\int_I |f - s| < \epsilon$ (Theorem 7.16); then $|\int f\sin(\alpha t+\beta)| \le \epsilon + |\int s\sin(\alpha t+\beta)| < 2\epsilon$ for large α. ∎

In particular $\hat{f}(n) \to 0$ as $|n| \to \infty$ (Theorem 9.21). The partial sums are integrals against the

Dirichlet kernel:

$$s_n(f;x)=\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)D_n(t)\,dt, \qquad D_n(t)=\sum_{|k|\le n} e^{ikt}=\frac{\sin(n+\frac{1}{2})t}{\sin\frac{t}{2}}, \qquad \frac{1}{2\pi}\int_{-\pi}^{\pi} D_n=1.$$

Since D_n is even, $s_n(f;x)=\frac{1}{2\pi}\int_0^{\pi}[f(x+t)+f(x-t)]D_n(t)\,dt$.

Theorem 9.4 (9.2.2, Riemann localization)

Let $0<\delta<\pi$ and $\int_{-\pi}^{\pi}|f|<\infty$. Then $\lim_n s_n(f;x)$ exists iff $\lim_n \frac{1}{\pi}\int_0^{\delta}[f(x+t)+f(x-t)]\frac{\sin(n+\frac{1}{2})t}{t}\,dt$ exists, and then they are equal. So convergence of the Fourier series at x depends only on f near x.

Proof. $\frac{1}{\sin(t/2)}-\frac{2}{t}$ is bounded on $(0,\pi]$, so by Theorem 9.3, $\frac{1}{2\pi}\int_0^{\pi}[f(x+t)+f(x-t)](\frac{1}{\sin(t/2)}-\frac{2}{t})\sin(n+\frac{1}{2})t\,dt\to 0$; and $\frac{2}{t}$ is bounded on $[\delta,\pi]$, so the integral over $[\delta,\pi]$ with kernel $\frac{2\sin(n+\frac{1}{2})t}{t}$ also tends to 0. ■

We use $\lim_{\lambda\to\infty}\int_0^{\delta}\frac{\sin\lambda t}{t}\,dt=\lim\int_0^{\lambda\delta}\frac{\sin u}{u}du=\frac{\pi}{2}$ (Example 8.2).

Lemma 9.5 (9.2.3)

If $g(0+)$ exists and $\int_0^{\delta}\frac{|g(t)-g(0+)|}{t}\,dt<\infty$, then $\lim_{\lambda\to\infty}\frac{2}{\pi}\int_0^{\delta}g(t)\frac{\sin\lambda t}{t}\,dt=g(0+)$.

Proof. Write $g(t)=g(0+)+(g(t)-g(0+))$. The first part gives $g(0+)\cdot\frac{2}{\pi}\cdot\frac{\pi}{2}$ in the limit; the second tends to 0 by Theorem 9.3 applied to the integrable function $(g(t)-g(0+))/t$. ■

Theorem 9.6 (9.2.4, Dini's test)

Let $f\in\mathcal{R}^1[-\pi,\pi]$ and $x\in[-\pi,\pi]$. Suppose (a) $f(x+)$ and $f(x-)$ exist, and (b) $\int_0^{\delta}\frac{|f(x+t)+f(x-t)-f(x+)-f(x-)|}{t}\,dt<\infty$. Then $s_n(f;x)\to\frac{1}{2}[f(x+)+f(x-)]$.

Proof. Apply Theorem 9.4 and Lemma 9.5 to $g(t)=\frac{1}{2}[f(x+t)+f(x-t)]$, with $\lambda=n+\frac{1}{2}$. ■

Corollary 9.7 (9.2.5)

If f satisfies a Lipschitz condition at x, $|f(x+t)-f(x)|\le Mt$ for small $|t|$, then $s_n(f;x)\to f(x)$.

Proof. The integrand in (b) is at most $2M$. ■

Corollary 9.8 (9.2.6)

If f is differentiable at x, then $s_n(f;x)\to f(x)$.

Proof. Differentiability at x implies $|f(x+t)-f(x)|\le(|f'(x)|+1)|t|$ for small t. ■

Lemma 9.9 (9.2.7)

If $g:[0,\delta]\to\mathbb{R}$ is of bounded variation, then $\lim_{\lambda\to\infty}\frac{2}{\pi}\int_0^{\delta}g(t)\frac{\sin\lambda t}{t}\,dt=g(0+)$.

Proof. By Jordan decomposition we may assume g increasing, and subtracting a constant, $g(0+) = 0$. Given ϵ, choose $h \in (0, \delta)$ with $g(h) < \epsilon$. On $[h, \delta]$, $g(t)/t$ is integrable, so that part tends to 0 by Theorem 9.3. On $(0, h]$, by the second mean value theorem (Corollary 5.25 of Volume I, applied on $[\eta, h]$ and letting $\eta \to 0$), $\int_0^h g(t)\frac{\sin \lambda t}{t}dt = g(h-)\int_c^h \frac{\sin \lambda t}{t}dt$ for some $c \in [0, h]$. Since $|\int_a^b \frac{\sin u}{u}du| \le K$ for all $0 \le a < b$ (a bounded quantity, as $\int_0^\infty \frac{\sin u}{u}$ converges), this part is at most ϵK. ∎

Theorem 9.10 (9.2.8, Dirichlet–Jordan)

If $f \in \mathcal{R}^1[-\pi, \pi]$ is of bounded variation on some interval $[x-\delta, x+\delta]$, then $s_n(f; x) \to \frac{1}{2}[f(x+) + f(x-)]$.

Proof. $g(t) = \frac{1}{2}[f(x+t)+f(x-t)]$ is of bounded variation on $[0, \delta]$ with $g(0+) = \frac{1}{2}[f(x+)+f(x-)]$; apply Theorem 9.4 and Lemma 9.9. ∎

Example 9.2

For $f(x) = x$ on $(-\pi, \pi)$ the series converges to x inside and to 0 at $\pm\pi$; at $x = \pi/2$ it gives $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$. Near the jump the partial sums overshoot by about 9% of the jump (Gibbs phenomenon). There exist continuous functions whose Fourier series diverge at a point (du Bois-Reymond).

9.3 Fejér's theorem

The **Cesàro means** are $\sigma_n(f; x) = \frac{1}{n+1}\sum_{k=0}^n s_k(f; x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)K_n(t)\,dt$ with the **Fejér kernel**

$$K_n(t) = \frac{1}{n+1}\sum_{k=0}^{n} D_k(t) = \frac{1}{n+1}\Big(\frac{\sin\frac{(n+1)t}{2}}{\sin\frac{t}{2}}\Big)^2.$$

It satisfies $K_n \ge 0$, $\frac{1}{2\pi}\int K_n = 1$, and $K_n(t) \le \frac{1}{(n+1)\sin^2(\delta/2)} \to 0$ uniformly on $\delta \le |t| \le \pi$. (The formula follows from $\sum_{k\le n}\sin(k+\frac{1}{2})t \cdot \sin\frac{t}{2} = \frac{1}{2}\sum(\cos kt - \cos(k+1)t) = \sin^2\frac{(n+1)t}{2}$.)

Theorem 9.11 (9.3.1, Fejér)

If $f \in \mathcal{R}^1[-\pi, \pi]$ and $f(x\pm)$ exist, then $\sigma_n(f; x) \to \frac{1}{2}[f(x+) + f(x-)]$. If f is continuous and 2π-periodic, $\sigma_n(f) \to f$ uniformly.

Proof. Since K_n is even, $\sigma_n(f; x) - s = \frac{1}{2\pi}\int_0^\pi [f(x+t)+f(x-t)-2s]K_n(t)\,dt$ with $s = \frac{1}{2}[f(x+)+f(x-)]$. Given ϵ, choose δ with $|f(x+t)+f(x-t)-2s| < \epsilon$ for $0 < t < \delta$. The integral over $(0, \delta)$ is at most $\epsilon \cdot \frac{1}{2\pi}\int_0^\pi K_n \le \epsilon/2$, and that over $[\delta, \pi]$ is at most $\frac{1}{(n+1)\sin^2(\delta/2)} \cdot \frac{1}{2\pi}\int |f(x+t)+f(x-t)-2s| \to 0$. For continuous periodic f, δ can be chosen uniformly in x and f is bounded, giving uniform convergence. ∎

So every continuous periodic function is a uniform limit of trigonometric polynomials (the trigonometric form of Weierstrass's theorem), and a continuous periodic function is determined by its Fourier coefficients.

Lemma 9.12 (9.3.2)

If $s_n \to s$, then $\sigma_n = \frac{1}{n+1}\sum_{k\le n} s_k \to s$.

Proof. We may assume $s = 0$. Given ϵ, choose N with $|s_n| < \epsilon$ for $n \geq N$. For $n > N$, $|\sigma_n| \leq \frac{1}{n+1}\sum_{k<N}|s_k| + \epsilon < 2\epsilon$ for large n. ■

Corollary 9.13 (9.3.3)

If the Fourier series of a continuous periodic f converges at x, its sum is $f(x)$.

Proof. By Lemma 9.12 the limit equals $\lim \sigma_n(f;x) = f(x)$ (Theorem 9.11). ■

Corollary 9.14 (9.3.4)

If $\int_{-\pi}^{\pi}|f|^2 < \infty$, then $\lim_n \|f - s_n(f)\|_2 = 0$.

Proof. Given ϵ, choose a continuous g vanishing near $\pm\pi$ (so continuous as a periodic function) with $\|f - g\|_2 < \epsilon$ (Theorem 7.14 and Lemma 7.13), and by Theorem 9.11 a trigonometric polynomial $T = \sigma_m(g)$ of degree m with $\|g - T\|_{\sup} < \epsilon$, so $\|g - T\|_2 < \epsilon$. For $n \geq m$, Theorem 9.1 gives $\|f - s_n(f)\|_2 \leq \|f - T\|_2 < 2\epsilon$. ■

Corollary 9.15 (9.3.5, Parseval's identity)

If $I = [a, a + 2\pi]$, $f \in \mathcal{R}^2(I)$ and $c_n = \frac{1}{2\pi}\int_I f(t)e^{-int}dt$, then $\frac{1}{2\pi}\int_I|f|^2 = \sum_{n\in\mathbb{Z}}|c_n|^2$; more generally $\frac{1}{2\pi}\int_I f\bar{g} = \sum c_n\overline{d_n}$ for $g \in \mathcal{R}^2(I)$ with coefficients d_n.

Proof. By periodicity the coefficients do not depend on a. From the proof of Theorem 9.1, $\|f\|_2^2 = \|f - s_n(f)\|_2^2 + \sum_{|k|\leq n}|c_k|^2$; let $n \to \infty$ (Corollary 9.14). The polarized form follows from $4\langle f, g\rangle = \sum_{k=0}^{3} i^k\|f + i^k g\|^2$. ■

Example 9.3

Parseval for $f(x) = x$ gives $\frac{1}{2\pi}\int_{-\pi}^{\pi} x^2dx = \frac{\pi^2}{3} = 2\sum_{n\geq 1}\frac{1}{n^2}$, again $\sum n^{-2} = \pi^2/6$.

9.4 Differentiation and integration of Fourier series

Theorem 9.16 (9.4.1, termwise integration)

If $f \in \mathcal{R}^2(I)$ has Fourier series $\sum\hat{f}(n)e^{int}$, then for $a, x \in I$, $\int_a^x f(t)\,dt = \sum_n \hat{f}(n)\int_a^x e^{int}dt$, the series converging (uniformly in x).

Proof. By Cauchy–Schwarz, $|\int_a^x(f - s_n(f))| \leq \sqrt{2\pi}\left(\int_I|f - s_n(f)|^2\right)^{1/2} \to 0$ by Corollary 9.14. ■

Note that the integrated series converges whether or not the Fourier series of f converges. If f is periodic and C^1, integration by parts gives $\widehat{f'}(n) = in\hat{f}(n)$; since $\sum|n\hat{f}(n)|^2 < \infty$, Cauchy–Schwarz gives $\sum|\hat{f}(n)| < \infty$, so the Fourier series converges uniformly (to f, by Corollary 9.13).

A sequence $a : \mathbb{Z} \to \mathbb{C}$ is **rapidly decreasing** if $|n|^k a(n)$ is bounded for every k.

Theorem 9.17 (9.4.2)

The Fourier coefficients of a C^∞ periodic function are rapidly decreasing. Conversely, if a is rapidly decreasing, $f(x) = \lim_n \sum_{|k|\le n} a(k)e^{ikx}$ defines a C^∞ periodic function with $\hat{f}(n) = a(n)$.

Proof. Repeated integration by parts gives $\widehat{f^{(k)}}(n) = (in)^k \hat{f}(n)$, so $|n^k \hat{f}(n)| \le \frac{1}{2\pi}\int |f^{(k)}|$. Conversely the series and all its term-by-term derivatives $\sum (ik)^j a(k)e^{ikx}$ converge uniformly by the M-test (as $\sum |k|^j |a(k)| < \infty$), so f is C^∞ (Theorem 6.4); integrating the uniformly convergent series against e^{-inx} term by term gives $\hat{f}(n) = a(n)$. ∎

Lemma 9.18 (9.4.3)

If $f : (a,b) \to \mathbb{R}$ is analytic and $K \subset (a,b)$ is compact, there are $C, R > 0$ with $|f^{(n)}(x)| \le Cn!R^n$ for $x \in K$, $n = 1, 2, \dots$.

Proof. Fix $c \in K$; near c, $f(x) = \sum_k a_k (x-c)^k$ with positive radius, so there are $\rho > 0$, M with $|a_k|\rho^k \le M$. For $|x - c| \le \rho/2$, termwise differentiation (Theorem 6.12) gives

$$|f^{(n)}(x)| \le \sum_{k \ge n} \frac{k!}{(k-n)!} |a_k| \Big(\frac{\rho}{2}\Big)^{k-n} \le M\rho^{-n} n! \sum_{j \ge 0} \binom{n+j}{n} 2^{-j} = 2M\, n! \Big(\frac{2}{\rho}\Big)^n.$$

Finitely many such neighbourhoods cover K; take the largest constants. ∎

Lemma 9.19 (9.4.4)

If $c \ge 0$, $R > 0$, $n \ge 1$ and $c \le k!\,(R/n)^k$ for all $k = 0, 1, 2, \dots$, then $c \le (R+1)e^{-n/(R+1)}$.

Proof. If $c > 0$, then $\frac{1}{k!}\big(\frac{n}{R+1}\big)^k \le \frac{1}{c}\big(\frac{R}{R+1}\big)^k$ for every k; summing, $e^{n/(R+1)} \le \frac{R+1}{c}$. ∎

Theorem 9.20 (9.4.5)

If $f : \mathbb{R} \to \mathbb{C}$ is periodic and analytic, then $\limsup_{|n|\to\infty} |\hat{f}(n)|^{1/|n|} < 1$.

Proof. By Lemma 9.18 on $K = [-\pi, \pi]$, $\|f^{(k)}\|_{\sup} \le Ck!R^k$ for $k \ge 1$ (and enlarging C, also for $k = 0$). Then $|\hat{f}(n)| = |\widehat{f^{(k)}}(n)|/|n|^k \le Ck!(R/|n|)^k$ for all k, so by Lemma 9.19, $|\hat{f}(n)| \le C(R+1)e^{-|n|/(R+1)}$, and $\limsup |\hat{f}(n)|^{1/|n|} \le e^{-1/(R+1)} < 1$. ∎

Conversely, such geometric decay of the coefficients makes f analytic (the series extends to a holomorphic function near the unit circle).

Theorem 9.21 (9.4.6)

If $\int_{-\pi}^{\pi} |f| < \infty$, then $\lim_{|n|\to\infty} \hat{f}(n) = 0$.

Proof. Theorem 9.3 applied to the real and imaginary parts of $\int f(t)(\cos nt - i \sin nt)\, dt$. ∎

The decay can be arbitrarily slow, but not every sequence tending to 0 is a sequence of Fourier coefficients:

Proposition 9.22 (9.4.7)

If $f \in \mathcal{R}^1[-\pi, \pi]$ is odd, then the partial sums of $\sum_{n\geq 1} \hat{f}(n)/n$ are bounded.

Proof. For odd f, $\hat{f}(n) = -\frac{i}{\pi}\int_0^\pi f(t)\sin nt\,dt$, so $\sum_{n\leq N}\frac{\hat{f}(n)}{n} = -\frac{i}{\pi}\int_0^\pi f(t)\sum_{n\leq N}\frac{\sin nt}{n}\,dt$. The sums $\sum_{n\leq N}\frac{\sin nt}{n}$ are bounded by $1+2\pi$ on $(0,\pi]$: the terms with $n \leq 1/t$ are at most t each, and for $n > 1/t$, summation by parts with (‡) of Chapter 6 gives at most $2\cdot t\cdot \pi/t$. Hence the partial sums are bounded by $\frac{1+2\pi}{\pi}\int_0^\pi |f|$. ■

So $\sum_{n\geq 2}\frac{\sin nx}{\log n}$, which converges everywhere (Theorem 6.20), is not the Fourier series of any $f \in \mathcal{R}^1$, since $\sum \frac{1}{n\log n} = \infty$.